SRM Institute of Science and Technology
Department of Mathematics
21MAB204T-Probability and Queueing Theory
Unit – III
Tutorial Sheet - 1

S. No.	Questions

Part – B (Each question carries 8 marks)

1 The joint probability density function of (X, Y) where X & Y are discrete is given in the following table. Are X and Y are independent?

X \ Y	0	1	2
0	0.1	0.04	0.06
1	0.2	0.08	0.12
2	0.2	0.08	0.12

2 The joint probability mass function of two random variables X and Y are given by $P(x,y)=\frac{1}{27}(2x+y), x=0,1,2, y=0,1,2$.

Find the conditional distribution of Y given X = 2.
Find the conditional distribution of X given Y = 2.

3 Three balls are drawn at R.V. without replacement from a box containing 2 white, 3 red and 4 black balls. If X denoted the number of white balls drawn and Y denotes the number of red balls drawn. Find the joint probability distribution of (X, Y)?

4 Given the following bivariate probability distribution:
(i) Compute the marginal distribution function of X and Y
(ii) The conditional distribution of X given Y = 2.

X \ Y	-1	0	1
0	1/15	2/15	1/15
1	3/15	2/15	1/15
2	2/15	1/15	2/15

5 The joint probability function X and Y are

X \ Y	-1	1
0	1/8	3/8
1	2/8	2/8

Find the correlation coefficient of X and Y.

Part – C (Each question carries 15 marks)

6 From the following distribution find (i) $P(X \le 1)$ (ii) $P(Y \le 3)$ (iii) $P(X \le 1, Y \le 3)$ (iv) $P(X \le 1 / Y \le 3)$ (v) $P(Y \le 3 / X \le 1)$.

X \ Y	1	2	3	4	5	6
0	0	0	1/32	2/32	2/32	3/32
1	1/16	1/16	1/8	1/8	1/8	1/8
2	1/32	1/32	1/64	1/64	0	2/64

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7 The joint probability mass function (X, Y) is given by (i) find the marginal distributions (ii) find the probability density of X+Y (iii) (iv) $P(X \le 1 / Y \le 3)$ (v) $P(Y \le 3 / X \le 1)$.

X \ Y	1	2	3
0	3k	6k	9k
1	5k	8k	11k
2	7k	10k	13k

8 Given joint pdf $f(X,Y) = \begin{cases} 8xy, 0 < x < 1, 0 < y < x \\ 0, \quad otherwise \end{cases}$. Find (i) the marginal density of X and Y. (ii) conditional density of X/(Y=y).